

* PCA on multivariate Data to reduce dimensionality and analyze structure.

10 rows — 6 columns
(People)

Age
BMI
Blood Pressure
Cholesterol
Daily walk steps
Hours of sleep

(A) — Pre Processing.

1. Standardize the dataset (mean=0, std dev=1 for each feature)

We use z-score formula to standardize

$$z = \frac{x - \mu}{s}$$

μ = column mean

s = sample std dev. (since $N < 30$)

$N = 10$

$$\mu = \frac{\sum x_i}{N} ; \quad s = \sqrt{\frac{\sum (x_i - \mu)^2}{N-1}} \Rightarrow s = \sqrt{s^2}$$

Mean:

$$\mu_{\text{Age}} = 25 + 32 + \dots + 31 / 10 = 352 / 10 = 35.2$$

$$\mu_{\text{BMI}} = 22 + 27 + \dots + 26 / 10 = 272 / 10 = 27.2$$

$$\mu_{\text{BP}} = 118 + 130 + \dots + 128 / 10 = 1351 / 10 = 135.1$$

$$\mu_{\text{chol}} = 180 + 210 + \dots + 185 / 10 = 2025 / 10 = 202.5$$

$$\mu_{\text{steps}} = 8000 + 6000 + \dots + 7100 / 10 = 64000 / 10 = 6400$$

$$\mu_{\text{sleep}} = 7.0 + 6.5 + \dots + 6.9 / 10 = 6.65$$

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Standard Deviation:

$$S = \sqrt{\frac{\sum (x_i - \mu)^2}{N-1}}$$

Age: $\mu_{Age} = 35.2$

$$x_i = \{ 25, 32, 45, 29, 35, 40, 27, 50, 38, 31 \}$$

$$x_i - \mu = [25 - 35.2, 32 - 35.2, \dots, 31 - 35.2]$$

$$x_i - \mu = [-10.2, -3.2, 9.8, -6.2, -0.2, 4.8, -8.2, 14.8, 2.8, -4.2]$$

$$(x_i - \mu)^2 = [104.04, 10.24, 96.04, 38.44, 0.04, 23.04, 67.24, 219.04, 7.84, 17.64]$$

$$\sum (x_i - \mu)^2 = 583.60$$

$$S = \sqrt{\frac{583.60}{9}} = \sqrt{64.8444} = 8.0526$$

BMI: $\mu_{BMI} = 27.2$

$$x_i = \{ 22, 27, 30, 24, 29, 31, 23, 32, 28, 26 \}$$

$$x_i - \mu = \{ -5.2, -0.2, 2.8, -3.2, 1.8, 3.8, -4.2, 4.8, 0.8, -1.2 \}$$

$$(x_i - \mu)^2 = \{ 27.04, 0.04, 7.84, 10.24, 3.24, 14.44, 17.64, 23.04, 0.64, 1.44 \}$$

$$\sum (x_i - \mu)^2 = 105.60$$

$$S = \sqrt{\frac{105.60}{9}} = \sqrt{11.7333} = 3.4254$$

Blood Pressure: $\mu_{BP} = 135.1$

$$x_i = \{ 118, 130, 145, 125, 140, 150, 120, 160, 135, 128 \}$$

$$x_i - \mu = \{ -22.5, 7.5, 17.5, -10.1, 4.9, 14.9, -15.1, 24.9, -0.1, -7.1 \}$$

$$(x_i - \mu)^2 = \{ 292.41, 26.01, 98.01, 102.01, 24.01, 222.01, 228.01, 620.01, 0.01, 50.41 \}$$

$$\sum (x_i - \mu)^2 = 1662.90$$

$$S = \sqrt{\frac{1662.90}{9}} = \sqrt{184.7667} = 13.5929$$

cholesterol: $\mu_{chole} = 202.5$

$$x_i = \{ 180, 210, 220, 190, 200, 230, 175, 240, 195, 185 \}$$

$$x_i - \mu = \{ -22.5, 7.5, 17.5, -12.5, -2.5, 27.5, -27.5, 37.5, -7.5, -17.5 \}$$

$$(x_i - \mu)^2 = \{ 506.25, 56.25, 306.25, 156.25, 6.25, 756.25, 756.25, 1406.25, 56.25, 306.25 \}$$

$$\sum (x_i - \mu)^2 = 4312.50$$

$$S_{chole} = \sqrt{\frac{4312.50}{9}} = \sqrt{479.1667} = 21.8899$$

walk steps: $\mu_{steps} = 6400$

$$x_i = \{ 8000, 6000, 5000, 7500, 6200, 4800, 8200, 4500, 6700, 7100 \}$$

$$x_i - \mu = \{ 1600, -400, -1400, 1100, -200, -1600, 1800, -1900, 300, 700 \}$$

$$(x_i - \mu)^2 = \{ 2560000, 160000, 1960000, 1210000, 40000, 2560000, 3240000, 3610000, 90000, 490000 \}$$

$$\sum (x_i - \mu)^2 = 15,920,000$$

$$S_{steps} = \sqrt{\frac{15,920,000}{9}} = \sqrt{1,768,888.889} = 1329.9958$$

Sleep: $\mu_{sleep} = 6.65$

$$x_i = \{ 7.0, 6.5, 6.0, 7.5, 6.8, 6.2, 7.2, 5.8, 6.6, 6.9 \}$$

$$x_i - \mu = \{ 0.35, -0.15, -0.65, 0.85, 0.15, -0.45, 0.55, -0.85, -0.05, 0.25 \}$$

$$(x_i - \mu)^2 = \{ 0.1225, 0.0225, 0.4225, 0.7225, 0.0225, 0.2025, 0.3025, 0.7225, 0.0025, 0.0625 \}$$

$$\sum (x_i - \mu)^2 = 2.6050$$

$$S_{sleep} = \sqrt{\frac{2.6050}{9}} = \sqrt{0.2894} = 0.5380$$

Standardize using $z\text{-score} = \frac{x - \mu}{s}$

Age: $\mu = 35.2$, $s = 8.0526$; $x_i = \{25, 32, 45, 29, 35, 40, 27, 50, 38, 31\}$

$$P_1 = \frac{25 - 35.2}{8.0526} = -1.2667$$

$$P_2 = \frac{32 - 35.2}{8.0526} = -0.397$$

$$P_3 = \frac{45 - 35.2}{8.0526} = 1.2170$$

$$P_4 = \frac{29 - 35.2}{8.0526} = -0.7699$$

$$P_5 = \frac{35 - 35.2}{8.0526} = -0.0248$$

$$P_6 = \frac{40 - 35.2}{8.0526} = 0.5961$$

$$P_7 = \frac{27 - 35.2}{8.0526} = -1.0183$$

$$P_8 = \frac{50 - 35.2}{8.0526} = 1.8379$$

$$P_9 = \frac{38 - 35.2}{8.0526} = 0.3477$$

$$P_{10} = \frac{31 - 35.2}{8.0526} = -0.5216$$

Old values = $\{25, 32, 45, 29, 35, 40, 27, 50, 38, 31\}$

New values = $\{-1.2667, -0.397, 1.217, -0.769, -0.0248, 0.5961, -1.0183, 1.8379, 0.347, -0.52\}$

$$\text{Mean} = \frac{\sum x_i (\text{New})}{N-1} = \frac{0}{9} = 0$$

$$\text{Std. Dev} = \sqrt{\frac{\sum (x_i - \mu)^2}{N-1}} = \sqrt{\frac{\sum x_i^2}{N-1}} \Rightarrow \text{New}$$

$$x_i^2 (\text{New}) = \{1.6045, 0.1579, 1.481, 0.592, 0.006, 0.3553, 1.036, 3.377, 0.120, 0.272\}$$

$$\sum x_i^2 (\text{New}) = 9.00014$$

$$s_{(\text{New})} = \sqrt{\frac{9.00014}{9}} \approx 1$$

Age $\rightarrow$ after standardized $\rightarrow$

Mean = 0
Std dev = 1

calculated
Manually.

Now data(Age) of 10 persons are centered around 0 and dispersed in 1 S.D. This data is now comparable.

Let's say

```
df = pandas.read_csv("data.csv")
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
df_scaled = scaler.fit_transform(df)
print(df_scaled)
```

python
code
for
whole
dataset

This will calculate the new values of whole data, just the way we calculated for AGE.

Person	Age	BMI	Blood Pressure	cholesterol	walkSteps	Hours of sleep
P ₁	-1.2667	-1.581	-1.2580	-1.0279	1.2030	0.6506
P ₂	-0.3974	-0.584	-0.3752	0.3426	-0.3008	-0.2788
P ₃	1.2170	0.8174	0.7283	0.7995	-1.0526	-1.2082
P ₄	-0.7699	-0.9342	-0.7430	-0.5710	0.8271	1.5779
P ₅	-0.0248	0.5255	0.3605	-0.1142	-0.1504	0.2788
P ₆	0.5961	1.1094	1.0962	1.2563	-1.2030	-0.8364
P ₇	-1.0183	-1.2261	-1.1109	-1.2563	1.3534	1.0223
P ₈	1.8379	1.4013	1.8318	1.7131	-1.4286	-1.5799
P ₉	0.3477	0.2335	-0.0074	-0.3426	0.2256	-0.829
P ₁₀	-0.5216	-0.3503	-0.5223	-0.7995	0.5263	0.4647

Transformed data

②. Compute and display covariance matrix of the standardize data.

Covariance
of
column 1
&
column 2



How column 1 co-vary w.r.t. column 2.

- to compare col 1 & col 2, data must be standardized. Hence we calculated.

- we can understand positive/Negative/No relationship b/w col 1 & col 2 based on co-variance values.

$$\text{Covariance}(X, Y) = \frac{1}{N-1} \sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})$$

∴ data is standardized ⇒ mean = 0

$$\boxed{\text{Cov}(X, Y) = \frac{1}{N-1} \sum_{i=1}^n (x_i) \cdot (y_i)}$$

Lets calculate covariance row-wise.

$$C_{11} = \frac{1}{9} \left[(-1.267)^2 + (-0.398)^2 + (1.217)^2 + (-0.770)^2 + (-0.025)^2 + (0.596)^2 + (-0.018)^2 + (1.838)^2 + (0.348)^2 + (-0.522)^2 \right]$$

$$C_{11} = \frac{1}{9} [9.001] = 1.00 \quad (\text{Age, Age})$$

$$C_{12} = \frac{1}{9} \left[(-1.267) \times (-1.518) + (-0.398) \times (-0.058) + (1.217) \times (0.414) + (-0.770) \times (0.924) + (-0.025) \times (0.526) + (0.596) \times (1.110) + (-1.018) \times (-1.26) + (1.838) \times (1.402) + (0.348) \times (0.234) + (-0.527) \times (0.350) \right]$$

$$= \frac{1}{9} [8.397] = 0.933 \quad (\text{Age, BMI})$$

$$C_{13} = \frac{1}{9} [(-1.267)(-1.258) + (-0.398)(-0.375) + (1.217)(0.728) + (-0.770)(0.743) + (-0.025)(0.860) + (0.596)(1.096) + (-1.018)(-1.111) + (1.838)(1.832) + (0.348)(-0.007) + (-0.522)(-0.522)]$$

$$= \frac{1}{9} [8.617] = 0.957 \quad [\text{Age, BP}]$$

$$C_{14} = \frac{1}{9} [(-1.267)(-1.023) + (-0.398)(0.341) + (1.217)(0.796) + (-0.770)(-0.568) + (-0.025)(-0.116) + (0.596)(1.251) + (-1.018)(-1.251) + (1.838)(1.705) + (0.348)(-0.341) + (0.522)(-0.796)]$$

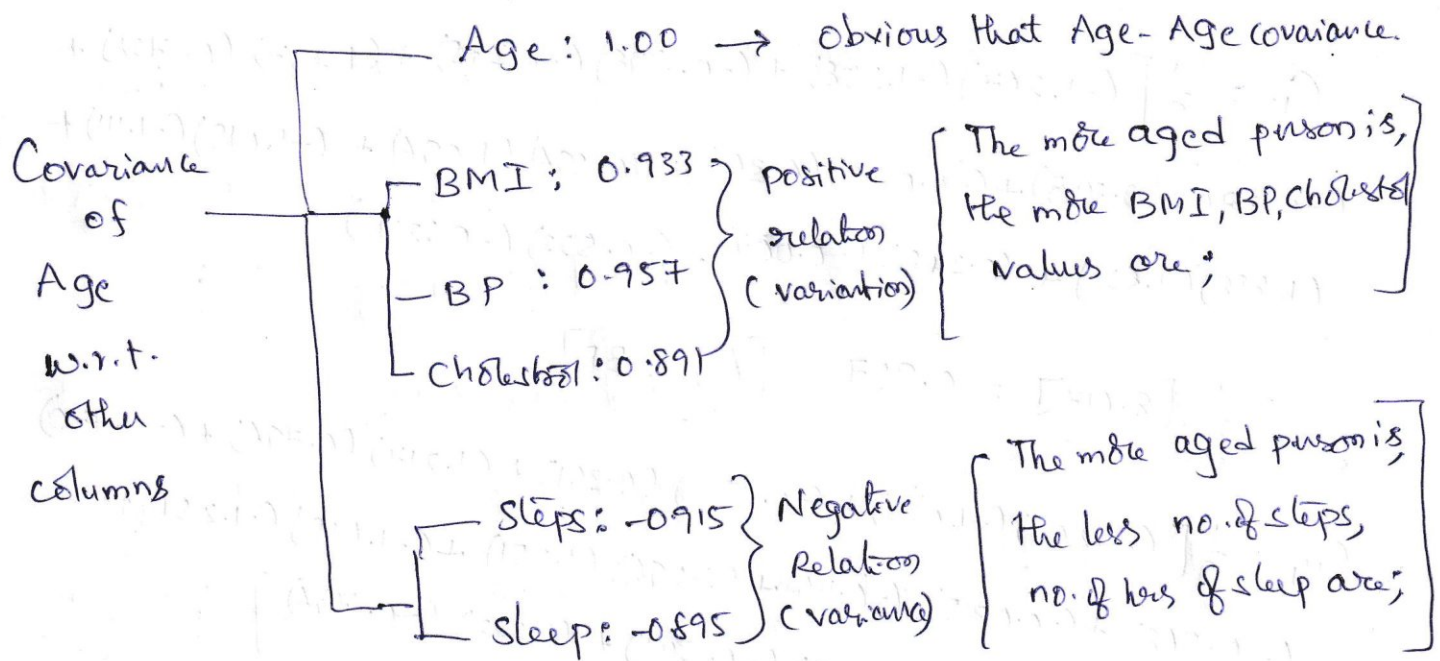
$$= \frac{1}{9} [8.019] = 0.891 \quad [\text{Age, cholesterol}]$$

$$C_{15} = \frac{1}{9} [(-1.267)(1.2030) + (-0.398)(-0.308) + (1.217)(-1.0526) + (-0.770)(0.827) + (-0.025)(-0.1504) + (0.596)(-1.2030) + (-1.018)(1.3539) + (1.838)(-1.4286) + (0.348)(0.225) + (0.522)(0.5263)]$$

$$= \frac{1}{9} [-8.236] = -0.915 \quad [\text{Age, sleep}]$$

$$C_{16} = \frac{1}{9} [(-1.267)(0.6506) + (-0.398)(-0.278) + (1.217)(-1.208) + (-0.770)(1.579) + (-0.025)(0.279) + (0.596)(-0.836) + (0.8030)(1.0223) + (1.838)(-1.579) + (0.348)(-0.093) + (0.522)(0.464)]$$

$$= \frac{1}{9} [-8.034] = -0.895 \quad [\text{Age, Hours of sleep}]$$



$$\begin{aligned} \text{Cov}(\text{col1}, \text{col1}) &= 1.00 \\ \text{Cov}(\text{col1}, \text{col2}) &= \text{Cov}(\text{col2}, \text{col1}) \end{aligned}$$

Let's calculate the other cells now —

$$C_{21} = C_{12} \quad [\text{Age, BMI}] = [\text{BMI, Age}]$$

$$C_{22} = 1.0 \quad [\text{BMI, BMI}]$$

$$\begin{aligned} C_{23} &= \frac{1}{9} [1.907 + 0.02 + 0.595 + 0.694 + 0.189 + 1.219 + 1.363 + 2.56 + \\ &\quad - 0.001 + 0.183] \\ &= \frac{1}{9} [8.736] = 0.970 \quad [\text{BMI, BP}] \end{aligned}$$

$$\begin{aligned} C_{24} &= \frac{1}{9} [1.560 + -0.020 + 0.653 + 0.534 - 0.060 + 1.39 + 1.540 + \\ &\quad 2.401 - 0.080 + 0.2808] \\ &= \frac{1}{9} [8.202] = 0.911 \quad [\text{BMI, cholesterol}] \end{aligned}$$

$$\begin{aligned} C_{25} &= \frac{1}{9} [-1.828 + 0.0175 - 0.860 - 0.772 - 0.0790 - 1.335 - 1.658 \\ &\quad - 2.001 + 0.0526 - 0.184] \\ &= \frac{1}{9} [-8.648] = -0.96093 \quad [\text{BMI, No. of steps}] \end{aligned}$$

$$C_{26} = \frac{1}{9} [-0.989 + 0.0162 - 0.987 - 1.473 + 0.143 - 0.927 - 1.254 - 2.212 - 0.0217 + 0.1626]$$

$$= \frac{1}{9} [-7.86814] = -0.874 \quad [\text{BMI, sleep}]$$

$C_3 = \text{Blood Pressure}$

$$C_{31} = C_{13} \quad [\text{Age, Blood Pressure}]$$

$$C_{32} = C_{23} \quad [\text{BMI, Blood Pressure}]$$

$$C_{33} = 1.00 \quad [\text{BP, BP}]$$

$$C_{34} = \frac{1}{9} [+1.292 - 0.128 + 0.582 + 0.424 - 0.0411 + 1.378 + 1.397 + 3.140 + 0.0025 + 0.417]$$

$$= \frac{1}{9} [8.46086] = 0.940 \quad [\text{BP, cholesterol}]$$

$$C_{35} = \frac{1}{9} [-1.512 + 0.112 - 0.766 - 0.614 - 0.0542 - 1.318 - 1.505 - 2.620 - 0.0017 - 0.274]$$

$$= \frac{1}{9} [-8.551560] = -0.950 \quad [\text{BP, steps}]$$

$$C_{36} = \frac{1}{9} [-0.817 + 0.1046 - 0.880 - 1.173 + 0.1005 - 0.916 - 1.135 - 2.893 + 0.0007 + 0.242]$$

$$= \frac{1}{9} [-7.8557] = -0.87286 \quad [\text{BP, sleep}]$$

C_4 : cholesterol

$$C_{41} = C_{14} \quad [\text{Age, cholesterol}]$$

$$C_{42} = C_{24} \quad [\text{BMI, cholesterol}]$$

$$C_{43} = C_{34} \quad [\text{BP, cholesterol}]$$

$$C_{44} = 1.00 \quad [\text{chole, chole}]$$

$$C_{45} = \frac{1}{9} \left[-1.236 - 0.103 - 0.842 - 0.472 + 0.0172 - 1.511 - 1.699 - 2.445 \right. \\ \left. - 0.0774 - 0.420 \right]$$

$$= \frac{1}{9} [-8.79327] = -0.977 \quad [\text{chole, steps}]$$

$$C_{46} = \frac{1}{9} \left[-0.668 - 0.0954 - 0.966 - 1.103 - 0.0318 - 1.051 - 1.284 - 2.704 \right. \\ \left. + 0.0318 - 0.371 \right]$$

$$= \frac{1}{9} [-8.0954] = -0.893 \quad [\text{chole, step}]$$

C_5 : steps

$$C_{51} = C_{15} \quad [\text{Age, steps}]$$

$$C_{52} = C_{25} \quad [\text{BMI, steps}]$$

$$C_{53} = C_{35} \quad [\text{BP, steps}]$$

$$C_{54} = C_{45} \quad [\text{chole, steps}]$$

$$C_{55} = 1.00 \quad [\text{steps, steps}]$$

$$C_{56} = \frac{1}{9} \left[0.783 + 0.0839 + 1.272 + 1.307 - 0.0419 + 1.006 + \right. \\ \left. 1.384 + 2.258 - 0.0209 + 0.2449 \right]$$

$$= \frac{1}{9} [8.27351] = 0.919 \quad [\text{steps, step}]$$

C_6 : sleep Hours

$C_{61} = C_{16}$ [Age, sleep]

$C_{62} = C_{26}$ [BMI, sleep]

$C_{63} = C_{36}$ [BP, sleep]

$C_{64} = C_{46}$ [chole, sleep]

$C_{65} = C_{56}$ [steps, sleep]

$C_{66} = 1.00$ [sleep, sleep]

Covariance matrix of standardized features

	Age	BMI	BP	chole	steps	sleep
Age	1.000	0.9329	0.957	0.895	-0.915	0.902
BMI	0.932	1.000	0.970	0.911	-0.960	-0.874
BP	0.957	0.970	1.000	0.940	-0.950	-0.882
chole	0.895	0.911	0.940	1.000	-0.977	-0.893
steps	-0.915	-0.960	-0.950	-0.977	1.000	0.919
sleep	0.902	-0.872	-0.872	-0.893	0.919	1.000

Python Code to do the above calculations

```
a = df['Age'].to_numpy()
b = df['BMI'].to_numpy()
cov = (a*b).sum() / (len(a)-1)
```

↑
2 columns - Covariance
manual calculation

```
cov = df.cov()
print(cov.round(3))
```

↑
Covariance matrix of
whole data frame
using built-in functions

[B]. PCA Computations:

3. Calculate the eigenvalues and eigen vectors of covariance matrix.

By using the previous covariance matrix C we will calculate Eigen values and Eigen vectors using power iteration method.

For matrix C we need to solve

$$Cx = \lambda x$$

$C \rightarrow$ Covariance matrix

$x \rightarrow$ Eigen vectors

$\lambda \rightarrow$ Eigen value

Largest Eigen values $\rightarrow$ keep the most variance (information)

Power Iteration process:

Let unit vector $V_0 = [1, 1, 1, 1, 1, 1]$

Normalize $V_0 = \frac{V_0}{\|V_0\|}$ where $\|V_0\| = \sqrt{V_1^2 + \dots + V_6^2}$

find the Eigen values $\lambda_k = V^T \cdot X \cdot C \cdot x$

Iteration 0: $V_0 = [1, 1, 1, 1, 1, 1]$

$$\|V_0\| = \sqrt{1^2 + 1^2 + \dots + 1^2} = \sqrt{6} = 2.449$$

$$V_0 = \frac{V_0}{\|V_0\|} = \frac{[1, 1, 1, 1, 1, 1]}{\sqrt{6}} = [0.40824, 0.40824, 0.40824, 0.40824, 0.40824, 0.40824]$$

$$= [0.40824, 0.40824, 0.40824, 0.40824, 0.40824, 0.40824]$$

Iteration 1: $V_1 = C \times V_0$

$$V_1[0] = (1.000 \times 0.4082) + (0.932 \times 0.4082) + 0.9570 \times 0.4082 + 0.8951 \times 0.4082 + (-0.9150) \times 0.4082 + (-0.9028 \times 0.4082)$$

$$V_1[0] = 0.803011$$

$$V_1[1] = 0.3808 + 0.4082 + 0.396 + 0.3719 - 0.3922 - 0.356848$$

$$V_1[\text{Age}] = 0.8081$$

$$V_1[2] = 0.390647 + 0.3962 + 0.4082 + 0.3837 - 0.3878 - 0.3563$$

$$V_1[\text{BP}] = 0.8346$$

$$V_1[3] = 0.365 + 0.3719 + 0.3837 + 0.4082 - 0.3988 - 0.3648$$

$$V_1[\text{Chol}] = 0.7656$$

$$V_1[4] = -0.3735 - 0.3922 - 0.3878 - 0.3988 + 0.4082 + 0.3752$$

$$V_1[\text{Steps}] = -0.7689$$

$$V_1[5] = -0.3685 - 0.3568 - 0.3563 - 0.3648 + 0.3752 + 0.4082$$

$$V_1[\text{Sleep}] = -0.6631$$

$$V_1 = [0.831, 0.808, 0.834, 0.765, -0.768, -0.663]^T$$

$$\|V_1\| = \sqrt{(0.831)^2 + (0.808)^2 + (0.834)^2 + (0.765)^2 + (-0.768)^2 + (-0.663)^2}$$

$$= \sqrt{3.6119} = 1.9005$$

$$V_1 = \frac{V_1}{\|V_1\|} = \frac{1}{1.9005} \begin{bmatrix} 0.8031 \\ 0.808 \\ 0.834 \\ 0.765 \\ -0.768 \\ -0.663 \end{bmatrix} = \begin{bmatrix} 0.422 \\ 0.425 \\ 0.431 \\ 0.402 \\ -0.404 \\ -0.348 \end{bmatrix}$$

(normalize)

Eigen
value V_1

$$\lambda_1 = V_1^T X C X V_1$$

$$C X V_1 = \begin{bmatrix} 2.285 \\ 2.306 \\ 2.324 \\ 2.288 \\ 2.331 \\ -2.217 \end{bmatrix}$$

$$\lambda_1 = V_1^T X (C X V_1) \rightarrow \text{dot product of } V_1^T \text{ and } (C X V_1)$$

$$= \begin{bmatrix} 0.422 \\ 0.425 \\ 0.438 \\ 0.402 \\ -0.404 \\ -0.348 \end{bmatrix} \begin{bmatrix} 2.28 \\ 2.306 \\ 2.324 \\ 2.288 \\ -2.331 \\ -2.217 \end{bmatrix} = \begin{aligned} & (0.422)(2.28) + (0.425)(2.306) + \\ & \dots + (-0.348)(-2.217) \end{aligned}$$

$$= 5.6064$$

$$\lambda_1 = 5.6064 \text{ (Eigenvalue 1)}$$

Similarly we calculate for all records (iterations)

Iteration 2:

$$V_2[0] = 0.422 + 0.396 + 0.4203 + 0.3605 + 0.370 + 0.315 = 2.285$$

$$V_2[1] = 0.394 + 0.425 + 0.426 + 0.367 + 0.388 + 0.305 = 2.306$$

$$V_2[2] = 2.324$$

$$V_2[3] = 2.288$$

$$V_2[4] = -2.331$$

$$V_2[5] = -2.217$$

$$V_2 = [2.285, 2.306, 2.324, 2.288, -2.331, -2.217]^T$$

$$\|V_2\| = \sqrt{31.537} = 5.615$$

$$V_2 = \frac{1}{5.615} [V_2] = \begin{bmatrix} 0.4069 \\ 0.4107 \\ 0.413 \\ 0.407 \\ -0.415 \\ -0.39 \end{bmatrix}$$

$$\text{Eigen value } \lambda_2 = V_2^T \cdot C V_2$$

$$C \cdot V_2 = \begin{bmatrix} 2.87 \\ 2.307 \\ 2.324 \\ -2.336 \\ -2.228 \end{bmatrix}$$

dot product

$$\lambda_2 = 5.6257$$

Similarly we calculate for all features using the same method.

$$C V_3 = \begin{bmatrix} 2.287 \\ 2.307 \\ 2.324 \\ 2.298 \\ -2.236 \\ -2.28 \end{bmatrix}$$

$$\|C V\| = 5.6257$$

$$V_3 = [0.4065, 0.4102, 0.4131, 0.407, -0.415, -0.396]^T$$

$$\lambda_3 = 5.625$$

We get the largest Eigen value $\lambda_1 = 5.6257$

Corresponding Eigen vector

$$V_1 = [0.4065, 0.4101, 0.4131, 0.4077, -0.415, -0.396]^T$$

$$PC1 \Rightarrow \text{Eigen value } 5.6257$$

$$\text{Eigen Explained Ratio } 0.9376 \Rightarrow 93.76\%$$

$$[PC1, PC2, PC3, PC4, PC5, PC6] \left. \vphantom{[PC1, PC2, PC3, PC4, PC5, PC6]} \right\} \text{6 standard features}$$

Eigen
values

$$[5.625, 0.158, 0.128, 0.062, 0.020, 0.003]$$

④ Rank the principal components based on the explained variance.

Eigen values $\lambda_i = \{5.625, 0.1589, 0.1289, 0.062, 0.0207, 0.0037\}$

total
variance $= \sum_{i=1}^6 \lambda_i$
 $= 6.0001 \Rightarrow 6.0$

Explained
variance
ratio $= \frac{\lambda_i}{\text{Total Variance}}$

$PC1 = \frac{5.625}{6.0} = 0.9375 = 93.75\% \text{ — Rank 1}$

$PC2 = 0.1589 / 6.0 = 0.0265 = 2.65\% \text{ — Rank 2}$

$PC3 = 0.1289 / 6.0 = 0.0215 = 2.15\% \text{ — Rank 3}$

$PC4 = 0.062 / 6.0 = 0.0104 = 1.04\% \text{ — Rank 4}$

$PC5 = 0.0207 / 6.0 = 0.0035 = 0.35\% \text{ — Rank 5}$

$PC6 = 0.0037 / 6.0 = 0.0006 = 0.06\% \text{ — Rank 6}$

Final Ranking: $PC1 > PC2 > PC3 > PC4 > PC5 > PC6$

$PC1 = 93.75\%$

$PC1 + PC2 = 96.41\%$

$PC1 + PC2 + PC3 = 98.559\%$

$PC1 + PC2 + PC3 + PC4 = 99.59\%$

$PC1 + PC2 + PC3 + PC4 + PC5 = 99.93\%$

$PC1 + PC2 + PC3 + PC4 + PC5 + PC6 = 100.00\%$

Cumulative
percentage

- ⑤ How many components are needed to explain at least 95% of the variance.

from the previous answer,

$$PC1 = 93.76\%$$

$$PC1 + PC2 = 96.41\% \Rightarrow \text{at least } 95\% \text{ variance.}$$

keeping $PC1, PC2$ makes data from 6D $\rightarrow$ 2D means 6 dimensionality to 2 dimensional while preserving 96.41% cumulative variance of original dataset.

C Projection and Visualization.

- ⑥ project data into the space of top m PC's from above answer.

Data projection $\rightarrow$ rotate the axes into new coordinates captures top 2 PC's

- Need to calculate 6D datapoints to 2D datapoints for given 10 peoples.

$$\boxed{T_i = ZW}$$

i^{th} person coordinates ($PC1$ & $PC2$)

Z - standardized matrix (10×6)

$W \in \mathbb{R}^{6 \times 2}$ (eigen vectors of 2 PCs)

$$PC1 \quad v_1 = [0.466, 0.4102, 0.4131, 0.4077, -0.4154, -0.3962]^T$$

$$PC2 \quad v_2 = [0.1193, 0.3918, 0.4070, -0.0885, 0.0489, 0.8102]^T$$

Each principal component is calculated using a dot product b/w standardized values and PCA loadings

$$PC1 = z_1 v_{11} + z_2 v_{12} + z_3 v_{13} + z_4 v_{14} + z_5 v_{15} + z_6 v_{16}$$

$z_i \rightarrow$ standardized values of features (Age, BMI, ... etc)

$v_{ij} \rightarrow$ weight of each feature in $PC1, PC2$

$$\begin{aligned} PC1_{P1} &= (-1.2667) \cdot (0.4066) + (-1.5181) \cdot (0.4102) + (-1.2580) \cdot (0.4131) \\ &\quad + (-1.0279) \cdot (0.4077) + (1.2030) \cdot (-0.4154) + (0.6506) \cdot (-0.3962) \\ &= -2.8340134 \end{aligned}$$

$$\begin{aligned} PC2_{P1} &= (-1.2667) \cdot 0.1193 + (-1.5181) \cdot 0.3918 + (-1.2580) \cdot 0.4070 + \\ &\quad + (-1.0279) \cdot (-0.0885) + (1.2030) \cdot 0.0489 + (0.6506) \cdot 0.8102 \\ &= -0.58106 \end{aligned}$$

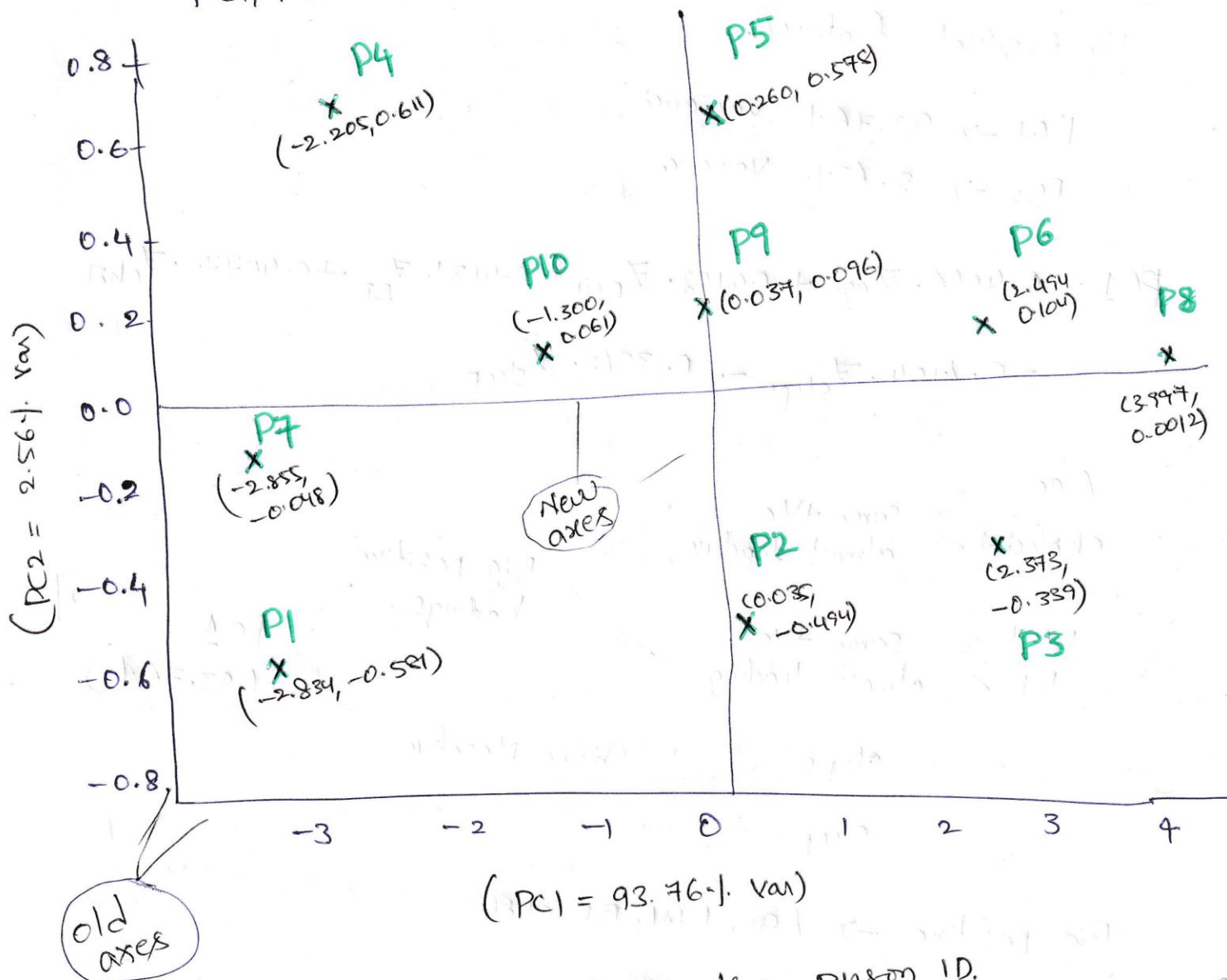
The same way we can calculate for every person $P_2 \dots P_{10}$.

Person	PC1	PC2
P ₁	-2.8340	-0.5810
P ₂	0.0346	-0.4939
P ₃	2.3729	-0.3392
P ₄	-2.2055	0.6107
P ₅	0.2599	0.5783
P ₆	2.4936	0.1043
P ₇	-2.8553	-0.0484
P ₈	3.9966	+0.0123
P ₉	0.0375	0.0960
P ₁₀	-1.3002	0.0609

2D
data
projection
for
10 people.

- ⑦ Visualize the data using a m -dimensional scatter plot of PCA,

$PC_1, PC_2 \rightarrow 2 = m \Rightarrow 10 \text{ data points}$



- ⑧ Label each point with corresponding person ID.

Let's reuse the above scatter-plot and label the data points of 10 persons in the above graph. That would make it more easy and interpretable.

please find in "Green Ink" persons IDs.

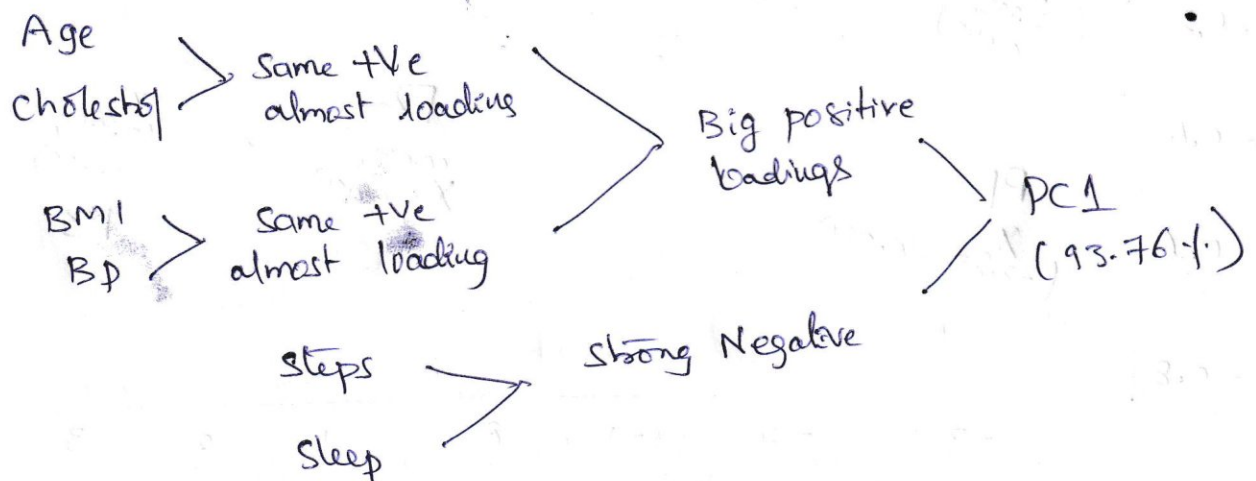
D) Analysis

9. Interpret the meaning of the first m PC's based on the original features.

PC1 $\rightarrow$ 93.76% variance

PC2 $\rightarrow$ 2.65% variance

$$PC1 = 0.4066 \cdot Z_{age} + 0.4102 \cdot Z_{BMI} + 0.4131 \cdot Z_{BP} + 0.4077 \cdot Z_{chol} \\ - 0.4154 \cdot Z_{steps} - 0.3962 \cdot Z_{sleep}$$



Big positive $\rightarrow$ Age, BMI, BP, chole

Big Negative $\rightarrow$ steps, sleep

Higher PC1 $\rightarrow$ older, higher BMI/BP/chole, few steps, sleep

Lower PC1 $\rightarrow$ younger, lower BMI/BP/chole, high steps, sleep

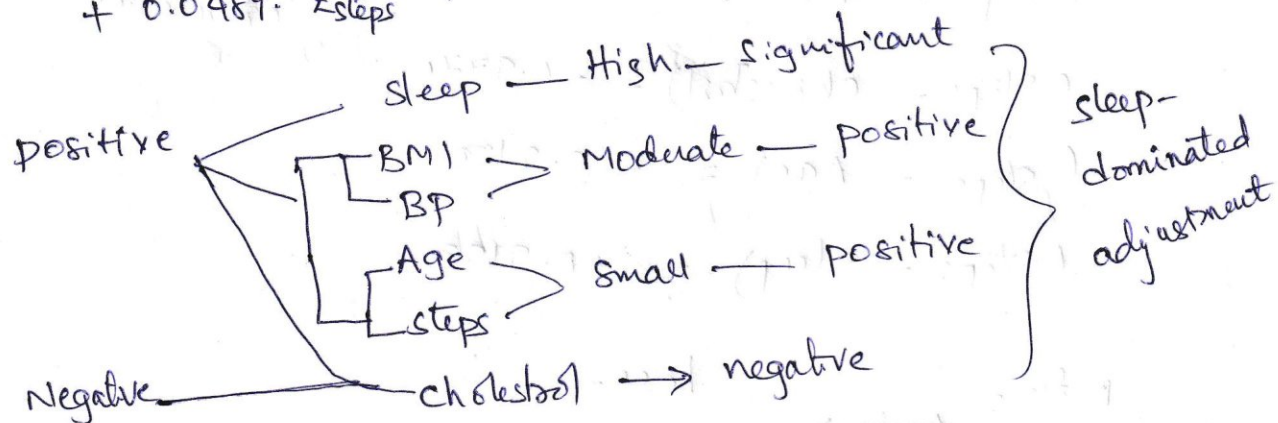
health
metrics

activity
health

PC1 $\rightarrow$ <health risk vs health Activity>

PC2 $\rightarrow$ explains 2.65% variance

$$PC2 = 0.1193 \times Z_{age} + 0.3918 \times Z_{BMI} + 0.4070 \times Z_{BP} - 0.0885 \times Z_{chol} \\ + 0.0489 \times Z_{steps} + 0.8102 \times Z_{sleep}$$



Let's take for example -

Person 8 $\rightarrow$ PC1 = 3.996 $\rightarrow$ older (50), high risk — health metrics
 PC2 = 0.0123 $\rightarrow$ less sleep effects (lowest from data) (5.8)

High — BMI, BP, chol
 32, 160, 240

(10) Are there any correlations between physical activity and health indicators visible in PCs?

physical activity $\rightarrow$ Daily walk steps

Health indicators $\rightarrow$ BMI, BP, cholesterol

$$\text{Correlation}(X, Y) = \frac{\text{Covariance}(X, Y)}{\sigma_X \sigma_Y}$$

for standardized data

From the covariance matrix
↓

$$(\text{Steps} - \text{BMI}) = -0.9609$$

$$(\text{Steps} - \text{BP}) = -0.9502$$

$$(\text{Steps} - \text{cholesterol}) = -0.9770$$

$$(\text{Steps} - \text{Age}) = -0.9150$$

$$(\text{Steps} - \text{sleep}) = +0.9193$$

Negative correlation

More - $\longleftrightarrow$ Less -
Steps $\longleftrightarrow$ BMI, BP, chole

More steps $\longleftrightarrow$ younger Age

More steps $\longleftrightarrow$ More sleep

From Eigenvectors for all PCs $\Rightarrow$ (Previous Answer)

Features $\Rightarrow$ Age, BMI, BP, chole, steps, sleep

$$\text{PC1} \Rightarrow 0.4066, 0.4102, 0.4131, 0.4077, -0.4154, -0.3962$$

$$\lambda_1 = 5.625, \sqrt{\lambda_1} = 2.3719$$

$$\text{Steps vs PC1} \Rightarrow -0.4154 \times 2.3719 = -0.9853 \Rightarrow \text{strong Negative}$$

$$\text{BMI vs PC1} = 0.4102 \times 2.3719 = 0.9729$$

$$\text{BP vs PC1} = 0.4131 \times 2.3719 = 0.9798$$

$$\text{chole vs PC1} = 0.4077 \times 2.3719 = 0.9670$$

Strong positive

PC1 $\Rightarrow$ strongly $\Rightarrow$ BMI, BP, chole $\Rightarrow$ lower activity
+ve

strongly $\Rightarrow$ steps, sleep $\Rightarrow$ higher activity
-ve

$$PC2 \Rightarrow vs \Rightarrow steps: \sqrt{\lambda_2} = \sqrt{0.1589} = 0.398$$

$$= 0.0489 \times 0.398 = 0.0195 \text{ (small)}$$

$$PC2 \text{ vs sleep} = 0.8102 \times 0.398 = 0.3230 \text{ (Moderate)}$$

$\therefore$ clear correlation b/w health indicators vs physical activity

proof 1: covariance/correlation matrix

proof 2: PC1, PC2 vs steps

reveal
strong
negative (inverse)
relationship b/w
physical activity vs
health markers

End